Mux4bit.v

Source code

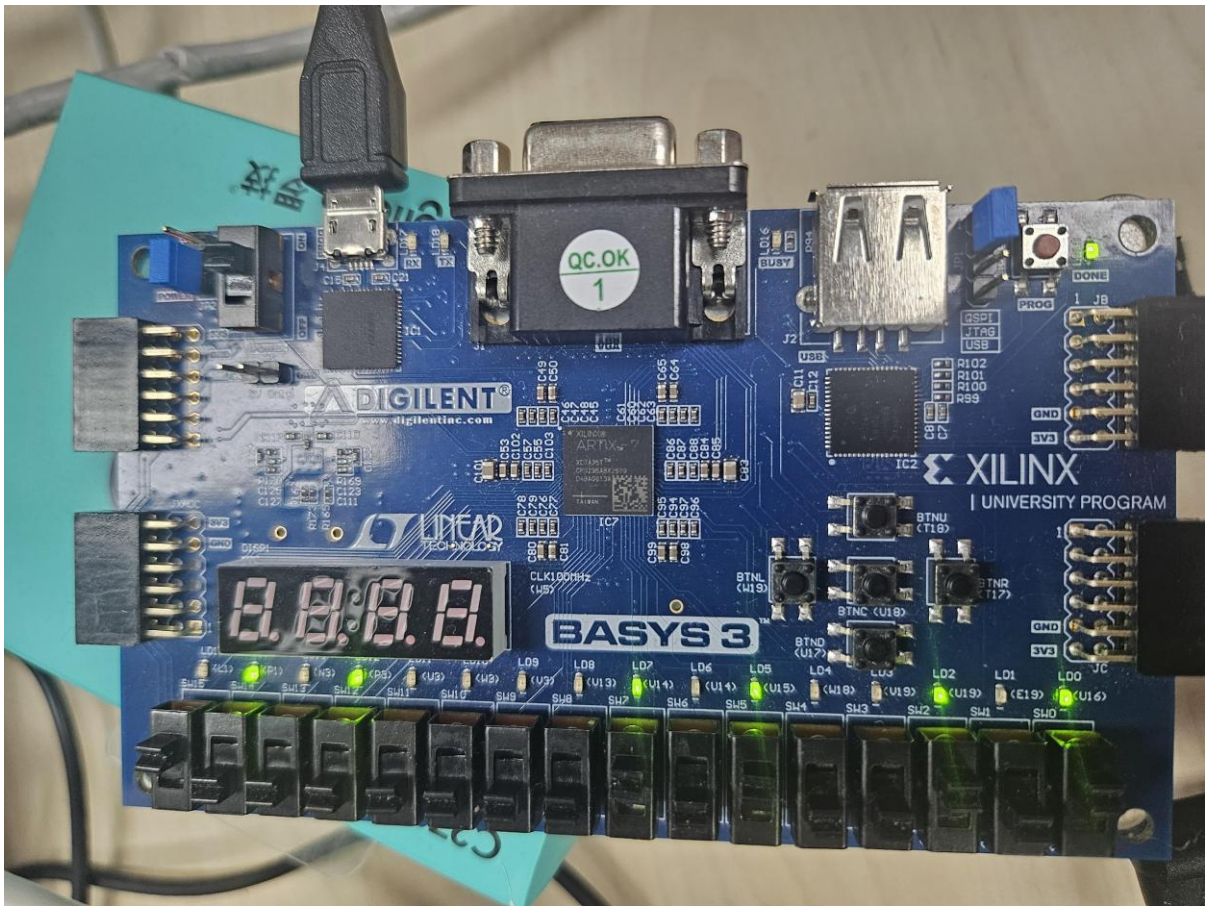
```
`timescale 1ns / 1ps


module mux4bit(a , b, sel, out , led);
    input [3:0] a, b;
    input sel;
    output[3:0] out;
    output[7:0] led;
    reg [3:0] out;

    assign led[3:0] = a;
    assign led[7:4] = b;

    always @(sel,a,b) begin
        case(sel)
        1'b1: out = a;
        1'b0: out = b;
        endcase
    end
endmodule
```

결과 사진



Tb_mux4bit.v

Source code

```
`timescale 1ns / 1ps


module tb_mux4bit();
    reg [3:0]a;
    reg [3:0]b;
    reg sel;
    wire [3:0] out;
    wire [7:0] led;

    mux4bit u_mux4bit(
```

```
        .a(a),
        .b(b),
        .sel(sel),
        .out(out),
        .led(led)
);

initial begin
        a = 4'b0000; b = 4'b0000; sel = 0;
        #10;
        $monitor("Time=%0t | sel=%b | a=%b | b=%b| out=%b", $time, sel, a, b, out);

        // a[0] 또는 b[3] 선택
        a = 4'b0001; b = 4'b1000;
        sel = 1; #10;   // out = 0001
        sel = 0; #10;   // out = 1000

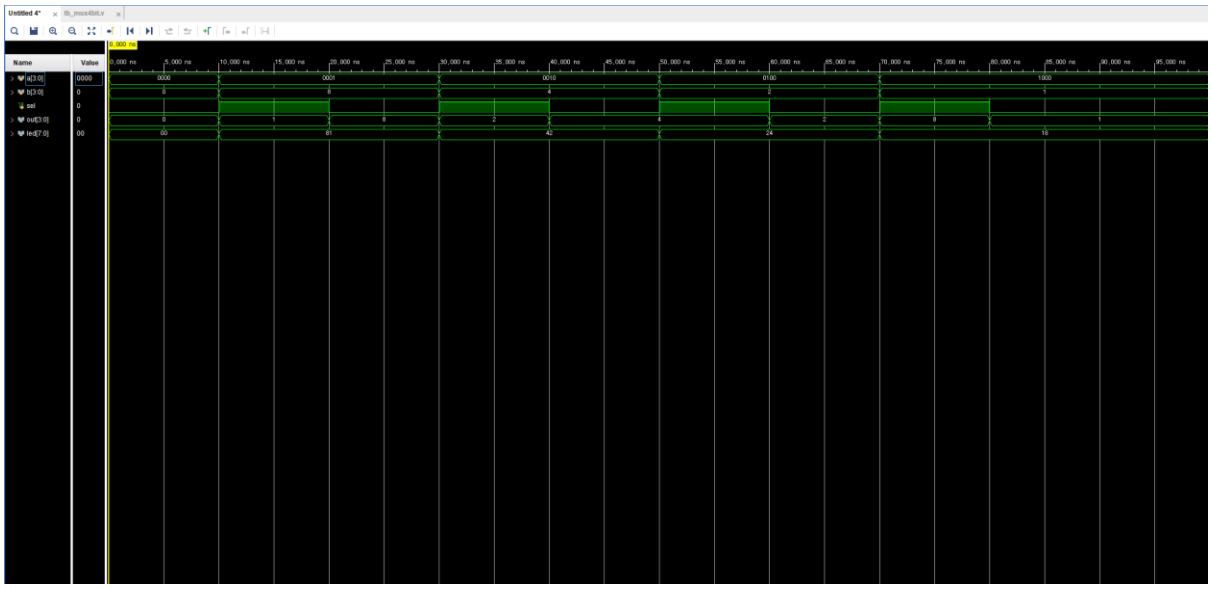
        // a[1] 또는 b[2] 선택
        a = 4'b0010; b = 4'b0100;
        sel = 1; #10;   // out = 0010
        sel = 0; #10;   // out = 0100

        // a[2] 또는 b[1] 선택
        a = 4'b0100; b = 4'b0010;
        sel = 1; #10;   // out = 0100
        sel = 0; #10;   // out = 0010
```

```
        // a[3] 또는 b[0] 선택
        a = 4'b1000; b = 4'b0001;
        sel = 1; #10;   // out = 1000
        sel = 0; #10;   // out = 0001

        #10;
        $finish;
    end
endmodule
```

실행결과



```
# run 1000ns
Time=10000 | sel=1 | a=0001 | b=1000| out=0001
Time=20000 | sel=0 | a=0001 | b=1000| out=1000
Time=30000 | sel=1 | a=0010 | b=0100| out=0010
Time=40000 | sel=0 | a=0010 | b=0100| out=0100
Time=50000 | sel=1 | a=0100 | b=0010| out=0100
Time=60000 | sel=0 | a=0100 | b=0010| out=0010
Time=70000 | sel=1 | a=1000 | b=0001| out=1000
Time=80000 | sel=0 | a=1000 | b=0001| out=0001
$finish called at time : 100 ns : File "D:/Project/fpga/ai2/04.mux4bit/04.mux4bit.srcs/sim_1/new/tb_mux4bit.v" Line 44
```